

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			As time increases temperature increases (1) at a constant rate / increase is linear (1) [Temperature <u>increase</u> [directly] proportional to time → (2)] Every 2 minutes temperature increase = 18 °C] or equiv (2) Temperature ∝ time → 1 mark		2		2		
	(b)	(i)		100 [°C]		1		1		
		(ii)		80 [°C] c.a.o.		1		1	1	
		(iii)		Substitution: $0.2 \times 4\,200 \times 80$ ecf (1) Thermal energy = 67 200 [J](1)	1	1		2	2	
	(c)			Water is changing state / becoming a gas / changing into steam Not: boiling or evaporating Accept answer in terms of molecules, e.g. particles getting further apart or bonds between {particles / molecules} broken			1	1		
				Question 5 total	1	5	1	7	3	0